

The Economics of PET Recycling

A Student Research Study for Brantford, Ontario

\$0.045	55%	54	3kg
Revenue per bottle	Profit margin	Households to break-even	CO2e saved per kg

Executive Summary

This research paper presents a comprehensive kg-based economic model for a door-to-door PET plastic bottle collection service operating in Brantford, Ontario. The study analyzes unit economics, break-even points, and environmental impact to demonstrate the feasibility of a student-run recycling initiative.

The Problem

Only 9% of all plastic ever produced has been recycled. In Brantford, households generate significant PET bottle waste that often ends up in landfills due to inconvenience of proper disposal. Current municipal recycling programs lack door-to-door collection for apartment dwellers and busy families.

Unit Economics

Our model generates \$0.045 revenue per bottle collected with a 55% profit margin. Each pickup collects 75-150 bottles per household. Weekly collection from 50 households generates approximately \$337-675 in weekly revenue. Costs include student wages, transportation, compression equipment, and storage.

Break-Even Analysis

The business reaches break-even at 54 registered households. At this threshold, weekly revenue covers all operational costs including student wages, fuel, and equipment. Beyond 54 households, each additional household contributes directly to profit and community impact.

Environmental Impact

Every kilogram of PET recycled saves 3 kg of CO₂ equivalent emissions. For a typical household generating 2-3 kg of PET bottles monthly, participation in Syscycl prevents 72-108 kg of CO₂e annually. At 100 households, this equals 7.2-10.8 tonnes of CO₂e prevented per year.

Methodology

Data was collected from Brantford municipal waste reports, interviews with local scrap dealers, EPA WARM model calculations, and comparable recycling programs across Ontario. Financial projections use conservative estimates with 20% buffer for operational variance.

Sources

EPA WARM Model (epa.gov/warm), Canadian Plastics Industry Association (plastics.ca), Brantford Municipal Waste Data (brantford.ca), Ontario Waste Management Association.